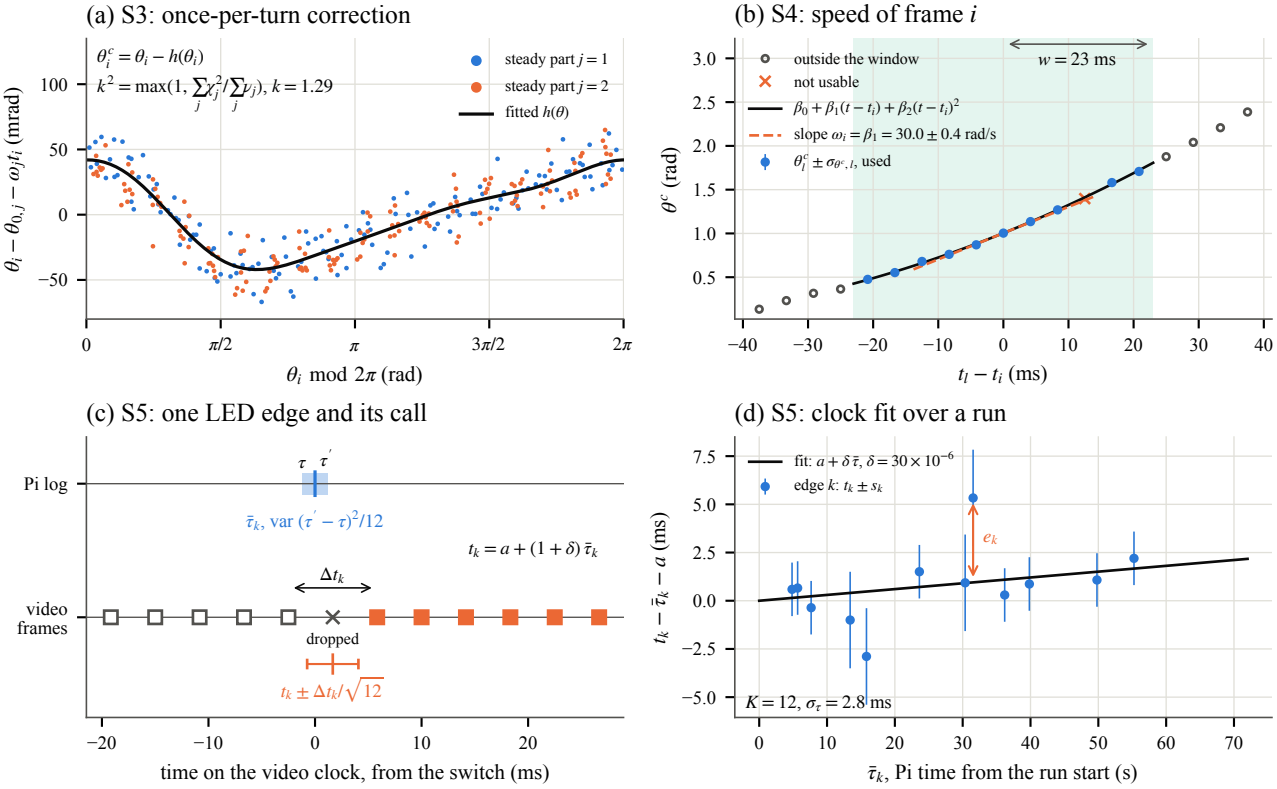

(a) S3: once-per-turn correction
$\theta_i^c = \theta_i - h(\theta_i)$
$k^2 = \max(1, \sum_j \chi_j^2 / \sum_j \nu_j)$, $k = 1.29$
steady part $j = 1$
steady part $j = 2$
fitted $h(\theta)$
$\theta_i - \theta_{0,j} - \omega_j t_i$ (mrad)
100
50
0
−50
0
$\pi/2$
π
$3\pi/2$
2π
θ_i mod 2π (rad)
(b) S4: speed of frame i
outside the window
not usable
$\beta_0 + \beta_1(t - t_i) + \beta_2(t - t_i)^2$
slope $\omega_i = \beta_1 = 30.0 \pm 0.4$ rad/s
$\theta_l^c \pm \sigma_{\theta^c, l}$, used
$w = 23$ ms
θ^c (rad)
3.0
2.5
2.0
1.5
1.0
0.5
−40
−30
−20
−10
0
10
20
30
40
$t_l - t_i$ (ms)
(c) S5: one LED edge and its call
Pi log
video frames
τ
τ'
$\bar{\tau}_k$, var $(\tau' - \tau)^2/12$
$t_k = a + (1 + \delta)\,\bar{\tau}_k$
Δt_k
dropped
$t_k \pm \Delta t_k/\sqrt{12}$
−20
−10
0
10
20
time on the video clock, from the switch (ms)
(d) S5: clock fit over a run
fit: $a + \delta\,\bar{\tau}$, $\delta = 30 \times 10^{-6}$
edge k: $t_k \pm s_k$
e_k
$K = 12$, $\sigma_\tau = 2.8$ ms
$t_k - \bar{\tau}_k - a$ (ms)
7.5
5.0
2.5
0.0
−2.5
−5.0
0
10
20
30
40
50
60
70
$\bar{\tau}_k$, Pi time from the run start (s)